

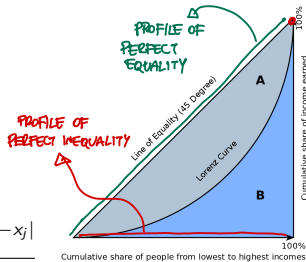
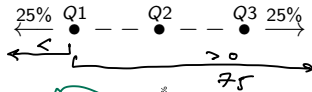
other SPREAD measures

ORDINAL VARIABLES (RANKED)

- INTERQUARTILE RANGE (IQR): $Q3 - Q1$

$\uparrow \text{IQR} \Rightarrow \uparrow \text{SPREAD}$

$\uparrow \text{GINI} \Rightarrow \uparrow \text{SPREAD}$



$$\text{GINI} = \frac{A}{A+B}$$

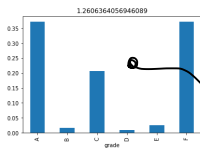
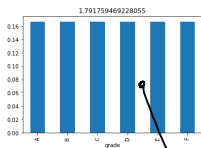
$$A=0 \rightarrow \text{GINI}=0$$

$$A \gg B \rightarrow \frac{A}{A+B} \rightarrow 1$$

$$\rightarrow \text{GINI}=1$$

- GINI COEFFICIENT: $G = \frac{\sum_{i=1,n} \sum_{j=1,n} |x_i - x_j|}{2n \sum_{i=1,n} x_i}$

- ENTROPY: $H = - \sum_{i=1,n} f_i \log(f_i)$ f_i FREQUENCY OF STATE i



$$0 < f_i < 1$$

SMALL H

$\uparrow H \Rightarrow \downarrow \text{SPREAD}$

$$1) f_i \rightarrow 1$$

$$\Rightarrow \log(f_i) \rightarrow 0$$

$$\Rightarrow f_i \log(f_i) \rightarrow 0$$

$$2) f_i \rightarrow 0$$

$$\Rightarrow f_i \log(f_i) \rightarrow 0$$

$$3) 0 < f_i < 1 \quad f_i \log(f_i) < 0$$

exercise 1.1

VARIANCE ($\sigma^2 = \frac{1}{n} \sum_{i=1,n} (x_i - \mu)^2$) has an alternative formula

UNCERTAIN VARIABLE X
 $E[X] = \text{MEAN}$

- $\sigma^2 = \frac{1}{n} \sum_{i=1,n} x_i^2 - \left(\frac{1}{n} \sum_{i=1,n} x_i\right)^2 = E[x^2] - E[x]^2$

where $E[\cdot]$ is the expected value (average)

- $\sigma^2 = \frac{1}{n} \sum_{i=1,n} (x_i - \mu)^2 = \frac{1}{n} \sum_{i=1,n} (x_i^2 + \mu^2 - 2x_i\mu) =$

$$\frac{1}{n} \sum_{i=1,n} x_i^2 + \frac{1}{n} \sum_{i=1,n} \mu^2 - \frac{2\mu}{n} \sum_{i=1,n} x_i = \frac{1}{n} \sum_{i=1,n} x_i^2 + \frac{1}{n} \sum_{i=1,n} \mu^2 - 2\mu^2 =$$

$$\frac{1}{n} \sum_{i=1,n} x_i^2 - \mu^2 - \mu^2 + \frac{1}{n} \sum_{i=1,n} \mu^2 = E[x^2] - \mu^2 - \mu^2 + \frac{1}{n} n \mu^2 =$$

$$E[x^2] - \mu^2 = E[x^2] - E[x]^2 = \sigma^2$$

alternative formula convenient for incremental computations!

exercise 1.2

